Lecture 2

1. **The first step in urine formation is:**

a) Glomerular filtration
b) Tubular reabsorption.
c) Tubular secretion.
d) Urine excretion.

Answer: A

2. **All the plasma undergoes filtration EXCEPT:**

a) Water.
b) Proteins.
c) Glucose.
d) Ions.

Answer: B

3. **Amount of glomerular fluid filtered per day is:**

a) 25 liters
b) 125 liters
c) 180 liters
d) 200 liters

Answer: C

4. **Which of the following is the main pressure that favors filtration in the glomerulus?**

a) Glomerular capillary hydrostatic pressure
b) Glomerular capillary colloid osmotic pressure
c) Bowman's capsule hydrostatic pressure
d) Bowman's capsule colloid osmotic Pressure

Answer: A

5. **Average filtration fraction is approximately:**

a) 20%
b) 25%
c) 80%
d) 60%

Answer: A

6. **Which change tends to increase GFR?**

a) Increased afferent arteriolar resistance

b) Decreased efferent arteriolar resistance
c) Increased glomerular capillary filtration coefficient
d) Increased Bowman's capsule hydrostatic pressure

Answer: C

7. **If the glomerular capillaries hydrostatic pressure is 50 mm Hg, the hydrostatic pressure in the Bowman's space is 12 mm Hg, the colloid osmotic pressure is 30 mm Hg, what is the net pressure driving glomerular filtration .....mm Hg**
   a) 8
   b) 32
   c) 48
   d) 60

Answer: A

8. **A negatively charged molecule is filtered with more difficulty compared to a positive one because:**
   a) Presence of negatively charged protein on the filtering membrane
   b) Negatively charged molecules are larger
   c) Positively charged molecules are smaller
   d) Urine is acidic

Answer: A

9. **Glomerular filtration rate is increased when**:
   a) Plasma oncotic pressure is increased
   b) Glomerular hydrostatic pressure is decreased
   c) Tubular hydrostatic pressure is increased
   d) Increased renal blood flow

Answer: D

10. **In renal disease albumin is the first protein to appear in urine because:**
    a) Of its high concentration in plasma
    b) Has molecular weight slightly lesser than other proteins
    c) High albumin: globulin ratio
    d) Tubular epithelial cells are sensitive to albumin

Answer: B

11. **Glomerular filtration rate increases by**:
    a) Afferent arteriolar constriction
    b) Efferent arteriolar constriction

c) Hypotension
d) Sympathetic stimulation

Answer: B

12. **Use the values below to answer the following question: Glomerular capillary hydrostatic pressure = 47 mmHg. Bowman capsule hydrostatic pressure = 10 mmHg at what value of glomerular capillary oncotic pressure would glomerular filtration stop ..... mmHg**
    a) 57
    b) 47
    c) 37
    d) 10
    e) 0

Answer: C

13. **Glomerular filtration rate and renal blood flow will both be increased if:**
    a) The afferent arteriole is dilated
    b) The efferent and afferent arterioles are both constricted
    c) Only the afferent arteriole is constricted
    d) Only the efferent arteriole is dilated

Answer: A

14. **Which of these factors oppose(s) filtration from the glomerulus?**
    a) plasma oncotic pressure
    b) hydrostatic pressure in glomerular (Bowman's) capsule
    c) plasma hydrostatic pressure
    d) both a and b

Answer: D

15. **Which of the following decreases filtration from the glomeruli**:
    a) an increase in the diameter of afferent arteriole.
    b) a mild constriction of the efferent arteriole.
    c) an increase in Bowman's capsule pressure.
    d) an increase in renal blood flow.
    e) an increase in filtration coefficient (K f).

Answer: C

16. **GFR is increased when:**
    a) afferent arteriole is constricted.

b) efferent arteriole is severely constricted.
c) efferent arteriole is mildly constricted.
d) efferent arteriole is dilated.
e) pressure in Bowman's capsule is increased.

Answer: C

17. **Which of the following causes an increase in GFR:**
    a) severe constriction of the efferent arterioles.
    b) constriction of the afferent arterioles.
    c) increased plasma osmotic pressure.
    d) increased Bowman's capsule osmotic pressure.
    e) decreased glomerular hydrostatic capillary pressure.

Answer: D

18. **The glomerular filtrate:**
    a) is different from plasma in its ionic composition.
    b) production rate is about 1 ml/min.
    c) is hypotonic compared to plasma.
    d) contains a small amount of albumin.

Answer: D